Run the following command to create a **resource controller**:

```
php artisan make:controller ProductController --api
```

This creates `app/Http/Controllers/ProductController.php` with **predefined CRUD methods**.

14.12.4.6 Step 6: Implement CRUD Methods in ProductController

Edit `app/Http/Controllers/ProductController.php` and update the methods:

```php
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;
use App\Models\Product;
use Illuminate\Http\JsonResponse;

class ProductController extends Controller
{
    // Get all products
    public function index(): JsonResponse
    {
        return response()->json(Product::all(), 200);
    }

    // Create a new product
    public function store(Request $request): JsonResponse
    {
        $validated = $request->validate([
            'name' => 'required|string|max:255',
            'description' => 'nullable|string',
            'price' => 'required|numeric|min:0',
            'stock' => 'required|integer|min:0',
        ]);

        $product = Product::create($validated);

        return response()->json($product, 201);
    }

    // Get a specific product
    public function show($id): JsonResponse
    {
        $product = Product::find($id);

        if (!$product) {
            return response()->json(['error' => 'Product not found'], 404);
        }

        return response()->json($product, 200);
    }

    // Update a product
    public function update(Request $request, $id): JsonResponse
    {
        $product = Product::find($id);
```